6

Nonlinear Filtering Theory

1. INTRODUCTION

In this chapter, we pursue the probabilistic approach to nonlinear filtering and prediction problems. As we concluded in Section 5.2, our approach begins with the determination of the equation of evolution for the conditional probability density function. We then develop equations of evolution for the conditional moments. Our interest lies especially in the conditional mean, which is the minimum variance estimate (Theorem 5.3). In addition, we obtain the equation of evolution for the conditional mode, which is the maximum likelihood (Bayesian) estimate (Section 5.3). Finally, the concept of limiting the memory (conditioning) of the filter is introduced, and a limited memory filter is developed.

The probabilistic approach to nonlinear filtering was pioneered by Stratonovich [13], and independently by Kushner [8, 9]. Much of the subsequent work in continuous filtering[1] is due to Kushner [10, 11]. Bucy [1] rederives Kushner's [9] result using stochastic calculus. This approach to continuous filtering is also taken by Wonham [14], Kashyap

[1] A new approach to continuous estimation has recently been developed by P. A. Frost [Nonlinear Estimation in Continuous Time Systems, Stanford Univ. Rept. SU-SEL-68-032, May 1968]. This approach, utilizing innovation processes, facilitates the treatment of nonlinear smoothing as well as filtering and prediction problems.

[7][2], and Fisher [2]. The results of Stratonovich [13] and Wonham [14] should be interpreted in the sense of Stratonovich.[3] Fisher treats a more general filtering problem than the one we have formulated (Section 5.1), but his analysis is completely formal. Mortensen [12] contains related theoretical work. The probabilistic approach to discrete filtering is developed in Ho and Lee [3]. Jazwinski [4, 5] treats the continuous-discrete problem.

2. CONTINUOUS-DISCRETE FILTERING

Our interest in filtering and prediction problems, in which the dynamics are continuous and observations discrete, stems from our interest in orbit determination. The dynamics of space vehicle trajectories are naturally continuous, and observations are usually taken at discrete time instants. This is especially true in the case of on-board manual navigation. Even in the case of radar tracking, where the observation rates can be quite high, limitations on computing speed and storage often dictate the discrete observation mode. Orbital dynamics can be effectively discretized, at least to first order. In highly nonlinear problems, however, the continuous dynamical model is more convenient.

Referring to Section 5.1, the dynamics will be described by the vector (Itô) stochastic differential equation (5.4):

$$dx_t = f(x_t, t)\,dt + G(x_t, t)\,d\beta_t, \qquad t \geqslant t_0. \tag{6.1}$$

Observations (5.2) y_k are taken at discrete time instants t_k :

$$y_k = h(x_k, t_k) + v_k; \qquad k = 1, 2,...; \qquad t_{k+1} > t_k \geqslant t_0. \tag{6.2}$$

The dimensions of the vectors and matrices appearing in (6.1) and (6.2), and the noise statistics, are defined in Section 5.1. The *a priori* data, $p(x_{t_0})$, is assumed given. We suppose that (6.1) satisfies the hypotheses

[2] Kashyap's report is apparently out of print and was not personally examined by this author.

[3] There is a problem in interpreting the results of Stratonovich [13] and Wonham [14] in the nonlinear case even if their stochastic integral is taken as a Stratonovich integral. This is because the Stratonovich integral (Section 4.6) is defined for explicit functions of η_t (Brownian motion), whereas [see (6.79)] in filtering problems the integrand is a functional on $[\eta_\tau : t_0 \leqslant \tau \leqslant t]$ through $\hat{h}_t$. In essence, the Stratonovich integral is not defined for the class of functions appearing in nonlinear filtering. Nevertheless, the pioneering work of Stratonovich stands as an inspiration to later investigators and should be recognized as such.

of Theorem 4.5, and that the function h is continuous in both arguments and bounded for each t_k wp 1.

Let

$$Y_\tau = \{y_l : t_l \leqslant \tau\}. \tag{6.3}$$

To solve the filtering problem, we must determine the evolution in time t of the conditional probability density function

$$p(x, t \mid Y_t). \tag{6.4}$$

3. EVOLUTION OF THE CONDITIONAL DENSITY (CONTINUOUS-DISCRETE)

Now in Section 4.9 we have already determined the evolution of the prior density $p(x, t)$, which is clearly the same as the conditional density (6.4) in the absence of observations. Therefore, *between* observations, $p(x, t \mid Y_t)$ satisfies Kolmogorov's forward equation (4.121)

$$dp(x, t \mid Y_t) = \mathscr{L}(p)\, dt, \qquad t_k \leqslant t < t_{k+1}, \tag{6.5}$$

where

$$\mathscr{L}(\cdot) = -\sum_{i=1}^{n} \frac{\partial(\cdot f_i)}{\partial x_i} + \frac{1}{2} \sum_{i,j=1}^{n} \frac{\partial^2(\cdot\,(GQG^{\mathrm{T}})_{ij})}{\partial x_i\, \partial x_j} \tag{6.6}$$

is the forward diffusion operator. The initial condition (at t_k) for (6.5) is

$$p(x, t_k \mid Y_{t_k}). \tag{6.7}$$

It remains to determine how p changes *at an observation* (at t_k); that is, to determine the relationship between $p(x, t_k \mid Y_{t_k})$ in (6.7) and

$$p(x, t_k \mid Y_{t_k^-}) \equiv p(x, t_k \mid Y_{t_{k-1}}). \tag{6.8}$$

Since

$$p(x, t_k \mid Y_{t_k}) = p(x, t_k \mid y_k, Y_{t_{k-1}}),$$

we have by Bayes' rule

$$p(x, t_k \mid Y_{t_k}) = \frac{p(y_k \mid x_k, Y_{t_{k-1}})\, p(x, t_k \mid Y_{t_{k-1}})}{p(y_k \mid Y_{t_{k-1}})}. \tag{6.9}$$

Now, since the noise $\{v_k\}$ in (6.2) is white,

$$p(y_k \mid x_k, Y_{t_{k-1}}) = p(y_k \mid x_k).$$

Similarly, it is easy to compute

$$p(y_k \mid Y_{t_{k-1}}) = \int p(y_k \mid x)\, p(x, t_k \mid Y_{t_{k-1}})\, dx.$$

Therefore, Eq. (6.9) becomes

$$p(x, t_k \mid Y_{t_k}) = \frac{p(y_k \mid x)\, p(x, t_k \mid Y_{t_k^-})}{\int p(y_k \mid \xi)\, p(\xi, t_k \mid Y_{t_k^-})\, d\xi}. \tag{6.10}$$

Once $p(y_k \mid x)$ is determined, (6.10) will provide the desired difference equation for the conditional density at an observation. By Theorem 2.7 and (6.2),

$$p(y_k \mid x) = p_{v_k}(y_k - h(x, t_k)), \tag{6.11}$$

and since $v_k \sim N(0, R_k)$,

$$\begin{aligned} p(y_k \mid x) = {} & (1/(2\pi)^{m/2} \mid R_k \mid^{1/2}) \\ & \times \exp\{-\tfrac{1}{2}[y_k - h(x, t_k)]^{\mathrm{T}} R_k^{-1}[y_k - h(x, t_k)]\}. \end{aligned} \tag{6.12}$$

We summarize these results in the following theorem.

Theorem 6.1 (Conditional Density for Continuous-Discrete Problems). *Let system* (6.1) *satisfy the hypotheses of Theorem* 4.5. *Suppose the prior density* $p(x, t)$ *for* (6.1) *exists and is once continuously differentiable with respect to* t *and twice with respect to* x. *Let* h *be continuous in both arguments and bounded for each* t_k wp 1. *Then, between observations, the conditional density* $p(x, t \mid Y_t)$ (6.4) *satisfies Kolmogorov's forward equation*

$$dp(x, t \mid Y_t) = \mathscr{L}(p)\, dt, \qquad t_k \leqslant t < t_{k+1}, \qquad p(x, t_0 \mid Y_{t_0}) = p(x_{t_0}), \tag{6.13}$$

where $\mathscr{L}$ (6.6) *is the forward diffusion operator. At an observation (at* t_k*), the conditional density satisfies the difference equation*

$$p(x, t_k \mid Y_{t_k}) = \frac{p(y_k \mid x)\, p(x, t_k \mid Y_{t_k^-})}{\int p(y_k \mid \xi)\, p(\xi, t_k \mid Y_{t_k^-})\, d\xi}, \tag{6.14}$$

where $p(y_k \mid x)$ *is given in* (6.12). *Prediction for* $t > \tau$, *based on* Y_τ (6.3), *is accomplished via Kolmogorov's forward equation*

$$dp(x, t \mid Y_\tau) = \mathscr{L}(p)\, dt, \qquad t > \tau, \tag{6.15}$$

where the initial condition $p(x, \tau \mid Y_\tau)$ *is the filtering solution.*

Note that the normalizing constant

$$1/(2\pi)^{m/2} \mid R_k \mid^{1/2}$$

in $p(y_k \mid x)$ is independent of x and therefore cancels between the numerator and denominator in (6.14). ($\exp\{-\frac{1}{2}y_k^{\mathrm{T}}R_k^{-1}y_k\}$ also cancels.) As a result of this, and since

$$\int p(\xi, t_k \mid Y_{t_k^-})\, d\xi = 1,$$

if $R_k^{-1} \equiv 0$, then

$$p(x, t_k \mid Y_{t_k}) = p(x, t_k \mid Y_{t_k^-}).$$

That is, if the observations are valueless (or there are no observations), then the conditional density does not change at the observation times. In that case the conditional density is identical to the prior density $p(x, t)$, whose evolution is then governed by Kolmogorov's forward equation (6.13).

One might in fact inquire as to the *value* of observations. Having processed the set of observations

$$\{y_1, \cdots, y_N\},$$

what have we learned, if anything, about the state x_{t_N} that is not contained in the prior data $p(x_{t_0})$? Such questions are clearly related to the properties of the system (6.1) and to the properties of the observation or measurement function h (6.2). The hypotheses we have imposed on the system and the function h merely guarantee the existence of the conditional probability density function. They say nothing about the actual value of the observations in improving our knowledge of the state.

Such questions as these are quite difficult, and no satisfactory theory is presently available in the general nonlinear problem. We shall return to this subject in Chapter 7, where we define and discuss the concepts of *observability* and *information* for linear filtering problems.

4. EVOLUTION OF MOMENTS (CONTINUOUS-DISCRETE)

Let $\varphi(x)$ be a twice continuously differentiable scalar function of the n-vector x. Define

$$\hat{\varphi}^\tau(x_t) = \mathscr{E}^\tau\{\varphi(x_t)\} \triangleq \mathscr{E}\{\varphi(x_t) \mid Y_\tau\} = \int \varphi(x)\, p(x, t \mid Y_\tau)\, dx, \qquad (6.16)$$

the expectation of φ using the conditional density $p(x, t \mid Y_\tau)$. We determine the evolution of $\hat{\varphi}^t(x_t)$ and $\hat{\varphi}^\tau(x_t)$ $(t > \tau)$ for the continuous-discrete filtering and prediction problems.

Now we have already determined the evolution of $\mathscr{E}\{\varphi(x_t)\}$ for scalar x_t in Examples 4.19 and 4.20, and the generalization to a vector x_t is straightforward. Since between observations both $p(x, t)$ and $p(x, t \mid Y_t)$ satisfy Kolmogorov's forward equation (6.13), the laws of evolution of $\mathscr{E}\{\varphi(x_t)\}$ and $\hat{\varphi}^t(x_t)$ are the same; specifically,

$$d\hat{\varphi}^t(x_t) = \mathscr{E}^t\{\varphi_x{}^{\mathrm{T}} f\}\, dt + \tfrac{1}{2} \operatorname{tr} \mathscr{E}^t\{GQG^{\mathrm{T}}\varphi_{xx}\}\, dt, \qquad t_k \leqslant t < t_{k+1}. \tag{6.17}$$

φ_x is the gradient, and φ_{xx} is the matrix of second partials. See Lemma 4.2.

We next compute the change in $\hat{\varphi}^t(x_t)$ *at an observation* (at t_k) using the difference equation (6.14) for the conditional density. Let

$$\hat{\varphi}^{t_k^+}(x_{t_k}) = \mathscr{E}^{t_k^+}\{\varphi(x_{t_k})\} = \int \varphi(x)\, p(x, t_k \mid Y_{t_k})\, dx, \tag{6.18}$$

$$\hat{\varphi}^{t_k^-}(x_{t_k}) = \mathscr{E}^{t_k^-}\{\varphi(x_{t_k})\} = \int \varphi(x)\, p(x, t_k) \mid Y_{t_k^-})\, dx. \tag{6.19}$$

Then, multiplying (6.14) by $\varphi(x)$ and integrating over x, we get

$$\hat{\varphi}^{t_k^+}(x_{t_k}) = \frac{\mathscr{E}^{t_k^-}\{\varphi(x_{t_k})\, p(y_k \mid x_{t_k})\}}{\mathscr{E}^{t_k^-}\{p(y_k \mid x_{t_k})\}}. \tag{6.20}$$

Note that y_k is fixed with respect to the expectation $\mathscr{E}^{t_k^-}$ in (6.20). Thus, the right-hand side of (6.20) is a function of y_k.

These results are summarized in the following lemma.

Lemma 6.1. *Hypotheses of Theorem* 6.1. *Let $\varphi(x)$ be a twice continuously differentiable scalar function of the n-vector x. Then, between observations $\hat{\varphi}^t(x_t)$ satisfies*

$$d\hat{\varphi}^t(x_t) = \mathscr{E}^t\{\varphi_x{}^{\mathrm{T}} f\}\, dt + \tfrac{1}{2} \operatorname{tr} \mathscr{E}^t\{GQG^{\mathrm{T}}\varphi_{xx}\}\, dt, \qquad t_k \leqslant t < t_{k+1}. \tag{6.21}$$

At an observation at t_k,

$$\hat{\varphi}^{t_k^+}(x_{t_k}) = \frac{\mathscr{E}^{t_k^-}\{\varphi(x_{t_k})\, p(y_k \mid x_{t_k})\}}{\mathscr{E}^{t_k}\{p(y_k \mid x_{t_k})\}}. \tag{6.22}$$

Fix τ. For $t \geqslant \tau$, $\hat{\varphi}^\tau(x_t)$ satisfies (6.21) *with $\mathscr{E}^t$ replaced by $\mathscr{E}^\tau$.*

From Lemma 6.1, we can determine the evolution of all the moments of the conditional density (6.4) for the continuous-discrete problem. We will be principally interested in the conditional mean

$$\hat{x}_t^{\tau} = \mathscr{E}^{\tau}\{x_t\} = \mathscr{E}\{x_t \mid Y_{\tau}\} \tag{6.23}$$

and the conditional covariance matrix

$$P_t^{\tau} = \mathscr{E}\{(x_t - \hat{x}_t^{\tau})(x_t - \hat{x}_t^{\tau})^{\mathrm{T}} \mid Y_{\tau}\} = \mathscr{E}^{\tau}\{x_t x_t^{\mathrm{T}}\} - \hat{x}_t^{\tau}\hat{x}_t^{\tau\mathrm{T}}. \tag{6.24}$$

The desired results clearly follow by setting $\varphi(x) = x_i$ and $\varphi(x) = x_i x_j$, respectively, in Lemma 6.1, where x_i is the ith component of the vector x. This was already done in Example 4.19 for (6.21) and x scalar.

Theorem 6.2 (Conditional Mean and Covariance Matrix for Continuous-Discrete Problems). *Hypotheses of Theorem* 6.1. *Between observations, the conditional mean and covariance matrix satisfy*

$$\begin{aligned} d\hat{x}_t^t/dt &= \hat{f}^t(x_t, t), \\ dP_t^t/dt &= (\mathscr{E}^t\{x_t f^{\mathrm{T}}\} - \hat{x}_t^t \hat{f}^{t\mathrm{T}}) + (\mathscr{E}^t\{f x_t^{\mathrm{T}}\} - \hat{f}^t \hat{x}_t^{t\mathrm{T}}) + \mathscr{E}^t\{GQG^{\mathrm{T}}\}, \\ & t_k \leqslant t < t_{k+1}, \end{aligned} \tag{6.25}$$

and, at an observation at t_k,

$$\begin{aligned} \hat{x}_{t_k}^{t_k+} &= \frac{\mathscr{E}^{t_k-}\{x_{t_k} p(y_k \mid x_{t_k})\}}{\mathscr{E}^{t_k-}\{p(y_k \mid x_{t_k})\}}, \\ P_{t_k}^{t_k+} &= \frac{\mathscr{E}^{t_k-}\{x_{t_k} x_{t_k}^{\mathrm{T}} p(y_k \mid x_{t_k})\}}{\mathscr{E}^{t_k-}\{p(y_k \mid x_{t_k})\}} - \hat{x}_{t_k}^{t_k+}\hat{x}_{t_k}^{t_k+\mathrm{T}}. \end{aligned} \tag{6.26}$$

$\hat{x}_t^{\tau}$ and P_t^{τ}, fixed $\tau < t$ (prediction based on Y_{τ}), satisfy (6.25) *with $\mathscr{E}^t$ replaced by $\mathscr{E}^{\tau}$.*

As was already noted in Example 4.19, Eqs. (6.25) are not ordinary differential equations. Neither are (6.26) ordinary difference equations. The right-hand sides of (6.25) and (6.26) involve expectations that require the whole conditional density for their evaluation. Thus the first two moments of the conditional density depend on all the other moments. Apparently, in order to obtain a computationally realizable and practical filter and predictor in the general nonlinear case, some approximation must be made. We shall consider such approximations in Chapter 9.

Equations (6.25) were specialized to the linear case in Example 4.19. See also Examples 4.15–4.18. In that case, the right-hand sides of (6.25) depend only on $\hat{x}_t{}^t$ and $P_t{}^t$. In fact, Eqs. (6.25) are uncoupled. Equations (6.26) will be specialized to the linear case in Chapter 7, where linear filters and predictors are developed.

5. EVOLUTION OF THE MODE (CONTINUOUS-DISCRETE)

The mode of the conditional density $p(x, t \mid Y_t)$ is an intuitively appealing estimate of the state x_t because of its maximum likelihood interpretation. According to the terminology introduced in Section 5.3, it is the maximum likelihood (Bayesian) estimate. In this section, we study the evolution of the conditional mode in continuous-discrete problems.

Consider first the evolution of the conditional mode *between* observations. The conditional density satisfies Kolmogorov's forward equation (6.13) of Theorem 6.1. That is,

$$\partial p/\partial t = \mathscr{L}(p). \tag{6.27}$$

If m_t is the mode of $p(x, t \mid Y_t)$, then we must have, by definition,

$$p_x(m, t \mid Y_t) \equiv 0, \tag{6.28}$$

identically in m and t, where p_x is the gradient (column vector) of p with respect to x. Consequently,

$$dp_x = (\partial p_x/\partial t)\, dt + p_{xx}\, dm = 0, \tag{6.29}$$

where p_{xx} is the matrix of second partials and where, henceforth, p and all its derivatives are evaluated at (m, t). We assume that

$$p_{xx}(m, t \mid Y_t) > 0$$

(is positive definite) at m, for all t, so that the mode is well defined, and that all the required partial derivatives of p exist.

Now, using (6.27),

$$\frac{\partial p_x}{\partial t} = \frac{\partial}{\partial x}\frac{\partial p}{\partial t} = (\mathscr{L}(p))_x,$$

and (6.29) produces

$$dm = -p_{xx}^{-1}(\mathscr{L}(p))_x\, dt, \tag{6.30}$$

which is the desired differential equation for the mode.

An explicit evaluation of the right-hand side of (6.30) reveals that it depends on pp_{xx}^{-1}, $p_{x_ix_j}$, and $p_{x_ix_jx_k}$ ($p_x \equiv 0$). If these derivatives were known, then (6.30) could be solved for m. Differential equations for these derivatives can be derived, but these depend on still higher-order derivatives. Apparently, as in the case of the conditional mean, approximations are required in order to obtain a realizable maximum likelihood filter (predictor).

Because in the linear-Gaussian case $-pp_{xx}^{-1}$ is the conditional covariance matrix, we shall derive a differential equation for it. Let

$$\Sigma_t = -pp_{xx}^{-1}. \tag{6.31}$$

Then

$$-d\Sigma_t = -pp_{xx}^{-1}\, dp_{xx}p_{xx}^{-1} + p_{xx}^{-1}\, dp. \tag{6.32}$$

Now

$$dp = p_x{}^{\mathrm{T}}\, dm + p_t\, dt = \mathscr{L}(p)\, dt, \tag{6.33}$$

since $p_x \equiv 0$. Proceeding,

$$(dp_{xx})_{ij} = \sum_k p_{x_ix_jx_k}\, dm_k + (\mathscr{L}(p))_{x_ix_j}\, dt,$$

and, substituting for dm from (6.30),

$$(dp_{xx})_{ij} = -\sum_k p_{x_ix_jx_k}[p_{xx}^{-1}(\mathscr{L}(p))_x]_k\, dt + (\mathscr{L}(p))_{x_ix_j}\, dt. \tag{6.34}$$

Substitution of (6.33) and (6.34) in (6.32) gives the desired equation.

There appears to be no simple means of computing the change in the mode at an observation. According to (6.14) of Theorem 6.1, the mode $m_{t_k^+}$ after the observation is the solution of the maximization problem

$$\max_x p(x, t_k \mid Y_{t_k}) = \max_x (1/c)\, p(y_k \mid x)\, p(x, t_k \mid Y_{t_k^-}), \tag{6.35}$$

where

$$c = \int p(y_k \mid \xi)\, p(\xi, t_k \mid Y_{t_k^-})\, d\xi$$

is independent of x. Therefore,

$$p_x(m_{t_k^+}, t_k \mid Y_{t_k}) = (1/c)[p_x(y_k \mid m_{t_k^+})\, p(m_{t_k^+}, t_k \mid Y_{t_k^-}) + p(y_k \mid m_{t_k^+})\, p_x(m_{t_k^+}, t_k \mid Y_{t_k^-})] = 0, \tag{6.36}$$

and, carrying out the differentiation of $p(y_k \mid x)$ and introducing the matrix notation

$$M = \left[\frac{\partial h_i}{\partial x_j}\right] \quad (m \times n),$$

we have

$$M^{\mathrm{T}}(m_{t_k^+})\, R_k^{-1}(y_k - h(m_{t_k^+}, t_k))\, p(m_{t_k^+}, t_k \mid Y_{t_k^-}) + p_x(m_{t_k^+}, t_k \mid Y_{t_k^-}) = 0. \quad (6.37)$$

Equation (6.37) is an n-vector nonlinear equation for $m_{t_k^+}$. It depends on the whole conditional density and its gradient. By differentiating (6.36) again and introducing the abbreviations

$$p(y_k \mid x) = p^v, \qquad p(x, t_k \mid Y_{t_k^-}) = p^x, \quad (6.38)$$

we get

$$\Sigma_{t^+}^{-1} = -(1/p^v p^x)(p_{xx}^v p^x + p_x{}^v p_x^{x\mathrm{T}} + p_x{}^x p_x^{v\mathrm{T}} + p^v p_{xx}^x). \quad (6.39)$$

Summarizing, we have

Theorem 6.3 (Conditional Mode for Continuous-Discrete Problems). *Hypotheses of Theorem* 6.1. *Let the conditional density be sufficiently differentiable, and suppose* $p_{xx} > 0$ *at the mode for all* t. *Then, between observations, the conditional mode satisfies*

$$dm_t = -p_{xx}^{-1}(\mathscr{L}(p))_x\, dt, \quad (6.40)$$

and $\Sigma_t\, (= -pp_{xx}^{-1})$ *satisfies*

$$-d\Sigma_t = -pp_{xx}^{-1}\, dp_{xx} p_{xx}^{-1} + p_{xx}^{-1}\, dp, \quad (6.41)$$

where dp *and* dp_{xx} *are given in* (6.33) *and* (6.34). *At an observation at* t_k, $m_{t_k^+}$ *satisfies*

$$M^{\mathrm{T}}(m_{t_k^+})\, R_k^{-1}(y_k - h(m_{t_k^+}, t_k))\, p(m_{t_k^+}, t_k \mid Y_{t_k^-}) + p_x(m_{t_k^+}, t_k \mid Y_{t_k^-}) = 0, \quad (6.42)$$

and

$$\Sigma_{t_k^+}^{-1} = -(1/p^v p^x)(p_{xx}^v p^x + p_x{}^v p_x^{x\mathrm{T}} + p_x{}^x p_x^{v\mathrm{T}} + p^v p_{xx}^x), \quad (6.43)$$

where p^v *and* p^x *are defined in* (6.38).

Example 6.1. Compute the right-hand sides of (6.40) and (6.41) in the scalar case ($n = m = 1$).

The results are

$$dm_t/dt = f - \Sigma_t f_{xx} - \frac{\sigma^2}{2}\frac{p_{xxx}}{p_{xx}} - \frac{3\sigma_x{}^2}{2} + \frac{1}{2}\Sigma_t\sigma^2_{xxx}, \tag{6.44}$$

$$\begin{aligned} d\Sigma_t/dt = {} & 2\Sigma_t f_x - \frac{\sigma^2}{2}\left(1 + \Sigma_t\frac{p_{xxxx}}{p_{xx}}\right) \\ & + \Sigma_t\frac{p_{xxx}}{p_{xx}}\left(\Sigma_t f_{xx} + \frac{\sigma^2}{2}\frac{p_{xxx}}{p_{xx}} + \frac{\sigma_x{}^2}{2} - \frac{1}{2}\Sigma_t\sigma^2_{xxx}\right) \\ & + \frac{1}{2}\Sigma_t{}^2(\sigma^2_{xxxx} - 2f_{xxx}) - \frac{5}{2}\Sigma_t\sigma^2_{xx}, \end{aligned} \tag{6.45}$$

where $\sigma^2 = GQG^{\mathrm{T}}$ and all the functions are evaluated at (m_t, t). Note that these equations depend on p_{xxx}/p_{xx} and p_{xxxx}/p_{xx}, and that equations for these quantities can be derived, or they can be approximated in some way.

Consider a special case in which p is symmetric $[p_{xxx}(m_t, t \mid Y_t) \equiv 0]$ and there is no system noise $(\sigma^2 = \sigma_x{}^2 = \sigma^2_{xx} = \sigma^2_{xxx} = \sigma^2_{xxxx} = 0)$. Then (6.44) and (6.45) reduce to

$$dm_t/dt = f - \Sigma_t f_{xx}, \tag{6.46}$$

$$d\Sigma_t/dt = 2\Sigma_t f_x - \Sigma_t{}^2 f_{xxx}. \tag{6.47}$$

Suppose f is quadratic and $f_{xx} > 0$. Then the equation for the conditional mean (4.164) is

$$d\hat{x}_t/dt = f(\hat{x}_t, t) + \tfrac{1}{2}P_t f_{xx}(\hat{x}_t, t), \tag{6.48}$$

where $P_t > 0$ is the conditional variance. (Recall that $\Sigma_t > 0$.) Comparing (6.46) with (6.48), we see that the evolution of the mode and the mean are radically different. For example, with $m_t = \hat{x}_t$ and $f(m_t, t) = 0$, $dm_t < 0$, whereas $d\hat{x}_t > 0$.

Now specialize (6.44) and (6.45) to the linear-Gaussian case;

$$f = F(t)\,x_t, \quad \sigma_x{}^2 = \sigma^2_{xx} = \sigma^2_{xxx} = \sigma^2_{xxxx} = p_{xxx} = 0, \quad p_{xxxx}/p_{xx} = -3\Sigma_t^{-1}.$$

We have

$$dm_t/dt = F(t)\,m_t, \tag{6.49}$$

$$d\Sigma_t/dt = 2\Sigma_t F(t) + \sigma^2, \tag{6.50}$$

which are in agreement with the equations for the mean and variance (4.138) previously obtained in Example 4.15. ▲

Example 6.2. In case observations are scalar[4] ($m = 1$),

$$p_x{}^v = p^v M^{\mathrm{T}}(m_{t_k^+})\, R_k^{-1}(y_k - h(m_{t_k^+}, t_k))$$

and

$$p_{xx}^v = p^v[M^{\mathrm{T}}MR_k^{-2}(y_k - h)^2 + h_{xx}R_k^{-1}(y_k - h) - M^{\mathrm{T}}MR_k^{-1}],$$

where

$$h_{xx} = \left[\frac{\partial h}{\partial x_i\, \partial x_j}\right] \qquad (n \times n),$$

and (6.43) becomes

$$\Sigma_{t_k^+}^{-1} = M^{\mathrm{T}}R_k^{-1}M - M^{\mathrm{T}}MR_k^{-2}(y_k - h)^2 - h_{xx}R_k^{-1}(y_k - h)$$
$$- (1/p^x)[M^{\mathrm{T}}p_x{}^{x\mathrm{T}}R_k^{-1}(y_k - h) + p_x{}^x MR_k^{-1}(y_k - h) + p_{xx}^x]. \quad (6.51)$$

If, in addition, p^x is assumed Gaussian with mean $m_{t_k^-}$ and covariance matrix $\Sigma_{t_k^-}$, then (6.42) and (6.51) become

$$-M^{\mathrm{T}}(m_{t_k^+})\, R_k^{-1}(y_k - h(m_{t_k^+}, t_k)) + \Sigma_{t_k^-}^{-1}(m_{t_k^+} - m_{t_k^-}) = 0, \quad (6.52)$$

$$\Sigma_{t_k^+}^{-1} = \Sigma_{t_k^-}^{-1} + M^{\mathrm{T}}(m_{t_k^+})\, R_k^{-1}M(m_{t_k^+}) - h_{xx}(m_{t_k^+})\, R_k^{-1}(y_k - h(m_{t_k^+}, t_k)). \quad (6.53)$$

In the linear-Gaussian (in general vector observation) case, $h = M(t_k)\, x_{t_k}$, these equations reduce to

$$m_{t_k^+} = (\Sigma_{t_k^-}^{-1} + M^{\mathrm{T}}R_k^{-1}M)^{-1}\,(M^{\mathrm{T}}R_k^{-1}y_k + \Sigma_{t_k^-}^{-1}m_{t_k^-}), \quad (6.54)$$

$$\Sigma_{t_k^+}^{-1} = \Sigma_{t_k^-}^{-1} + M^{\mathrm{T}}R_k^{-1}M. \quad (6.55)$$

Equations (6.49) and (6.50), together with (6.54) and (6.55), constitute the maximum likelihood (Bayesian) filter and predictor in the linear case for continuous-discrete problems. This is clearly the same as the minimum variance filter. In the scalar case ($n = m = 1$), (6.54) and (6.55) can be easily manipulated into the following form:

$$m_{t_k^+} = m_{t_k^-} + \frac{\Sigma_{t_k^-}M}{M^2\Sigma_{t_k^-} + R}(y_k - Mm_{t_k^-}), \quad (6.56)$$

$$\Sigma_{t_k^+} = \Sigma_{t_k^-} - \frac{\Sigma_{t_k^-}^2 M^2}{M^2\Sigma_{t_k^-} + R}. \quad (6.57)$$

The results of this example will be discussed in some detail in Chapters 7 and 9. ▲

[4] This special case is considered only for notational convenience.

6. DISCRETE FILTERING

Our discrete dynamical system is described by the vector difference equation (5.1),

$$x_{k+1} = \varphi(x_k, t_{k+1}, t_k) + \Gamma(x_k, t_k) w_{k+1}, \qquad k = 0, 1,..., \tag{6.58}$$

and the discrete observations are given by (5.2),

$$y_k = h(x_k, t_k) + v_k, \qquad k = 1,...; \qquad t_{k+1} > t_k. \tag{6.59}$$

The statistics of the noise inputs w_k and v_k, and the dimensions of the vectors and matrices appearing in the foregoing, are given in Section 5.1. $p(x_0)$ is, of course, given.

Let

$$Y_l = \{y_1, ..., y_l\}. \tag{6.60}$$

We determine the evolution in k of the conditional densities

$$p(x_k, t_k \mid Y_k)$$

and

$$p(x_k, t_k \mid Y_l),$$

fixed $l \leqslant k$. These densities provide, respectively, solutions to the discrete filtering and prediction problems.

Theorem 6.4 (Conditional Density for Discrete Problems). *Assume that the required densities exist. $p(x_k, t_k \mid Y_k)$ satisfies the integral difference equation*

$$p(x_{k+1}, t_{k+1} \mid Y_{k+1}) = \frac{p(y_{k+1} \mid x_{k+1}) \int p(x_{k+1} \mid x_k)\, p(x_k, t_k \mid Y_k)\, dx_k}{\iint p(y_{k+1} \mid x_{k+1})\, p(x_{k+1} \mid x_k)\, p(x_k, t_k \mid Y_k)\, dx_k\, dx_{k+1}}, \tag{6.61}$$

where $p(y_k \mid x_k)$ is given in (6.12), *and $p(x_{k+1} \mid x_k)$ is the transition probability density of* (6.58). *Prediction for $k \geqslant l$ (l fixed) is accomplished via*

$$p(x_{k+1}, t_{k+1} \mid Y_l) = \int p(x_{k+1} \mid x_k)\, p(x_k, t_k \mid Y_l)\, dx_k. \tag{6.62}$$

PROOF: Actually, these results have already been derived. Equation (6.62) is just the Chapman–Kolmogorov equation (3.100). Equation (6.14) of Theorem 6.1 can be rewritten as

$$p(x_{k+1}, t_{k+1} \mid Y_{k+1}) = \frac{p(y_{k+1} \mid x_{k+1})\, p(x_{k+1}, t_{k+1} \mid Y_k)}{\int p(y_{k+1} \mid x_{k+1})\, p(x_{k+1}, t_{k+1} \mid Y_k)\, dx_{k+1}}. \tag{6.63}$$

Setting $l = k$ in (6.62) and substituting into (6.63) gives (6.61). ■

It remains to specify the transition probability densities $p(x_{k+1} \mid x_k)$. Unfortunately, these cannot be written conveniently in general; the various special cases are discussed in Section 3.9. As a result, we shall not pursue the nonlinear discrete filtering and prediction problem any further, but we shall return to Theorem 6.4 in Chapter 7, where we develop the linear case.

7. CONTINUOUS FILTERING

We now turn to continuous filtering and prediction problems in which the state evolves according to the vector (Itô) stochastic differential equation (5.4)

$$dx_t = f(x_t, t)\,dt + G(x_t, t)\,d\beta_t, \qquad t \geqslant t_0, \tag{6.64}$$

and the observed process is the solution of the vector Itô equation (5.5)

$$dz_t = h(x_t, t)\,dt + d\eta_t, \qquad t \geqslant t_0. \tag{6.65}$$

The dimensions of (6.64) and (6.65) are given in Section 5.1. $\{\beta_t, t \geqslant t_0\}$ and $\{\eta_t, t \geqslant t_0\}$ are independent Brownian motion processes with $\mathcal{E}\{d\beta_t\, d\beta_t^{\mathrm{T}}\} = Q(t)\,dt$, $\mathcal{E}\{d\eta_t\, d\eta_t^{\mathrm{T}}\} = R(t)\,dt$, $R(t) > 0$. $p(x_{t_0})$ is given, and x_{t_0} is independent of $\{\beta_t\}$ and $\{\eta_t\}$.

Let

$$Y_\tau = \{z_s, t_0 \leqslant s \leqslant \tau\}. \tag{6.66}$$

In this section, we characterize the conditional density

$$p(x, t \mid Y_t). \tag{6.67}$$

Subsequent sections will be devoted to the evolution of the conditional density, its moments, and its mode.

Partition the interval $[t_0, t]$,

$$t_0 = \tau_0 < \tau_1 < \cdots < \tau_n = t,$$

and let $\rho = \max_i (\tau_{i+1} - \tau_i)$. We shall construct the conditional density

$$p(x, t \mid z_{\tau_0}, \ldots, z_{\tau_n})$$

and formally *define*

$$p(x, t \mid Y_t) = \operatorname*{l.i.m.}_{\substack{\rho \to 0 \\ n \to \infty}} p(x, t \mid z_{\tau_0}, \ldots, z_{\tau_n}). \tag{6.68}$$

See Section 3.5, particularly (3.82).

Lemma 6.2. *Let* (6.64) *satisfy the hypotheses of Theorem* 4.5, *and let the prior density* $p(x, t)$ *be sufficiently differentiable. Let* h *be continuous in both arguments, and*

$$\int_{t_0}^{t} h^{\mathrm{T}}(x_\tau, \tau)\, R^{-1}(\tau)\, h(x_\tau, \tau)\, d\tau < \infty \tag{6.69}$$

for every $t > t_0$ wp 1 *and in mean square, so that*

$$\int_{t_0}^{t} \mathscr{E}\{h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau\}\, d\tau < \infty \text{ [5]} \tag{6.70}$$

for every $t > t_0$. *Suppose the expectation below exists. Then*

$$p(x, t \mid Y_t) = \frac{\mathscr{E}^t \left\{ \exp\left[-\frac{1}{2} \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau\, d\tau + \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, dz_\tau \right] \middle|\, x_t \right\} p(x, t)}{\mathscr{E}^t \left\{ \exp\left[-\frac{1}{2} \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau\, d\tau + \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, dz_\tau \right] \right\}}, \tag{6.71}$$

where the expectation $\mathscr{E}^t$ *is over the* $\{x_\tau, \tau \in [t_0, t]\}$ *process with* Y_t *fixed.*

The result (6.71), a characterization of the conditional density, is stated by Bucy [1] and proved in Mortensen [12]. We will not prove it here, since the mathematics involved is beyond the scope of this book. For example, the expectation in (6.71) is over functions and requires the tools of function space integration. Rather, using the mean square definition of conditional density (6.68), and considering the expectation in (6.71) as a formal average, we give a formal demonstration of the lemma.

A rigorous proof of (6.71) apparently requires that h be uniformly bounded [12]. This, for example, excludes the linear case which is studied in Chapter 7. However, in Chapter 7, we essentially need only Lemma 6.3 which has been proved by Kushner [10] without this boundedness condition. Furthermore, the continuous linear filter is derived independently in the examples of Chapter 7.

Demonstration. Note that the hypotheses in (6.69) and (6.70) ensure that the integrals in (6.71) exist. The second integral is

$$\int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, dz_\tau = \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau\, d\tau + \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, d\eta_\tau,$$

where the last integral is an Itô integral.

[5] (6.69) in mean square implies (6.70) by Theorem 3.7 and its Corollary 1.

Now, by Bayes' rule,

$$p(x, t \mid z_{\tau_0}, \ldots, z_{\tau_n}) = \frac{p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_t)\, p(x, t)}{p(z_{\tau_0}, \ldots, z_{\tau_n})}. \tag{6.72}$$

Also,

$$p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_t) = \mathscr{E}^t\{p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_{\tau_0}, \ldots, x_{\tau_{n-1}}, x_t) \mid x_t\} \tag{6.73}$$

and

$$p(z_{\tau_0}, \ldots, z_{\tau_n}) = \mathscr{E}^t\{p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_t)\}, \tag{6.74}$$

and so we need only compute

$$p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_{\tau_0}, \ldots, x_{\tau_n}).$$

Integrating (6.65), and assuming a fine partition, we have

$$z_{\tau_{k+1}} - z_{\tau_k} = h(x_{\tau_k}, \tau_k)(\tau_{k+1} - \tau_k) + (\eta_{\tau_{k+1}} - \eta_{\tau_k}) + o(\rho). \tag{6.75}$$

We shall henceforth ignore the $o(\rho)$ terms. In view of (6.75),

$$\begin{aligned} p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_{\tau_0}, \ldots, x_{\tau_n}) &= p(z_{\tau_n} \mid z_{\tau_{n-1}}, \ldots, z_{\tau_0}, x_{\tau_0}, \ldots, x_{\tau_n}) \\ &\quad \cdot p(z_{\tau_{n-1}}, \ldots, z_{\tau_0} \mid x_{\tau_0}, \ldots, x_{\tau_n}) \\ &= p(z_{\tau_n} \mid z_{\tau_{n-1}}, x_{\tau_0}, \ldots, x_{\tau_n})\, p(z_{\tau_{n-1}}, \ldots, z_{\tau_0} \mid x_{\tau_0}, \ldots, x_{\tau_n}) \\ &\vdots \\ &= p(z_{\tau_n} \mid z_{\tau_{n-1}}, x_{\tau_0}, \ldots, x_{\tau_n})\, p(z_{\tau_{n-1}} \mid z_{\tau_{n-2}}, x_{\tau_0}, \ldots, x_{\tau_n}) \\ &\quad \cdots p(z_{\tau_0}), \end{aligned} \tag{6.76}$$

since $\{\eta_\tau\}$ has independent increments. Therefore, using (6.75) and Theorem 2.7, we get

$$\begin{aligned} &p(z_{\tau_0}, \ldots, z_{\tau_n} \mid x_{\tau_0}, \ldots, x_{\tau_n}) \\ &= p(z_{\tau_0})\, c(\rho) \exp\Big[-\frac{1}{2}\sum_{k=0}^{n-1} (\tau_{k+1} - \tau_k)^{-1}\{(z_{\tau_{k+1}} - z_{\tau_k}) - h_{\tau_k}(\tau_{k+1} - \tau_k)\}^{\mathrm{T}} \\ &\quad \times R^{-1}(\tau_k)\{(z_{\tau_{k+1}} - z_{\tau_k}) - h_{\tau_k}(\tau_{k+1} - \tau_k)\}\Big]. \end{aligned} \tag{6.77}$$

When we substitute (6.77) into (6.73) and (6.74), and subsequently into (6.72), $p(z_{\tau_0})$, $c(\rho)$, and the

$$(z_{\tau_{k+1}} - z_{\tau_k})^{\mathrm{T}} R^{-1}(\tau_k)(z_{\tau_{k+1}} - z_{\tau_k})$$

terms cancel between numerator and denominator, and we get

$$p(x, t \mid z_{\tau_0}, \ldots, z_{\tau_n})$$

$$= K\mathscr{E}^t \left\{ \exp \left[-\frac{1}{2} \sum_{k=0}^{n-1} h_{\tau_k}^{\mathrm{T}} R^{-1}(\tau_k)\, h_{\tau_k}(\tau_{k+1} - \tau_k) \right. \right.$$

$$\left. \left. + \sum_{k=0}^{n-1} h_{\tau_k}^{\mathrm{T}} R^{-1}(\tau_k)(z_{\tau_{k+1}} - z_{\tau_k}) \right] \right| x_t \Bigg\} p(x, t), \tag{6.78}$$

where K is a normalization constant that we shall determine later. Now

$$\underset{\substack{\rho \to 0 \\ n \to \infty}}{\text{l.i.m.}} \sum_{k=0}^{n-1} h_{\tau_k}^{\mathrm{T}} R^{-1}(\tau_k)\, h_{\tau_k}(\tau_{k+1} - \tau_k) = \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau \, d\tau,$$

and

$$\underset{\substack{\rho \to 0 \\ n \to \infty}}{\text{l.i.m.}} \sum_{k=0}^{n-1} h_{\tau_k}^{\mathrm{T}} R^{-1}(\tau_k)(z_{\tau_{k+1}} - z_{\tau_k}) = \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, dz_\tau .$$

Therefore, by Example 3.11,

$$\underset{\substack{\rho \to 0 \\ n \to \infty}}{\text{l.i.m.}} p(x, t \mid z_{\tau_0}, \ldots, z_{\tau_n})$$

$$= K\mathscr{E}^t \left\{ \exp \left[-\frac{1}{2} \int_{t_0}^{t} h_\tau^{\mathrm{T}} R^{-1}(\tau)\, h_\tau \, d\tau + \int_{t_0}^{t} h_\tau^{T} R^{-1}(\tau)\, dz_\tau \right] \right| x_t \Bigg\} p(x, t).$$

$1/K$ is clearly the denominator in (6.71), and this completes the demonstration.

8. EVOLUTION OF THE CONDITIONAL DENSITY (CONTINUOUS)

Using Lemma 6.2 and the Itô stochastic calculus (Section 4.5), we now develop the equation of evolution for $p(x, t \mid Y_t)$ (6.67).

Theorem 6.5 (Conditional Density for Continuous Problems). *Hypotheses of Lemma* 6.2. *Then the conditional density* $p(x, t \mid Y_t)$ *satisfies Kushner's equation*

$$dp = \mathscr{L}(p)\, dt + (h_t - \hat{h}_t)^{\mathrm{T}} R^{-1}(t)(dz_t - \hat{h}_t\, dt)\, p, \qquad t \geqslant t_0 , \tag{6.79}$$

where

$$\hat{h}_t = \mathscr{E}^t\{h(x_t, t)\} = \int h(x, t)\, p(x, t \mid Y_t)\, dx.$$

Equation (6.79) was first obtained by Kushner [8, 9] and later by Bucy [1]. Our derivation follows that of Bucy; Kushner's derivation is developed in Example 6.3. The Stratonovich analog of (6.79) is derived in [13] (see footnote 3 in this chapter).

Kushner's equation is a stochastic partial differential equation for the conditional density. No theory for equations of this type is currently available, and (6.79) can only be interpreted formally. Using (6.79) we shall, in the next two sections, develop equations of evolution for the moments and the mode of p. Fortunately, the equations for the moments have been rigorously established by Kushner [10].

With valueless or no observations [$R^{-1}(t) \equiv 0$], Kushner's equation reduces to Kolmogorov's forward equation (6.13). Thus, as in the continuous-discrete problem, prediction is accomplished via Kolmogorov's forward equation.

PROOF: Defining

$$\begin{aligned} d\zeta_t &= -\tfrac{1}{2}h_t^{\mathrm{T}}R^{-1}(t)\,h_t\,dt + h_t^{\mathrm{T}}R^{-1}(t)\,dz_t \\ &= \tfrac{1}{2}h_t^{\mathrm{T}}R^{-1}(t)\,h_t\,dt + h_t^{\mathrm{T}}R^{-1}(t)\,d\eta_t\,, \end{aligned} \tag{6.80}$$

Eq. (6.71) of Lemma 6.2 becomes

$$p(x, t \mid Y_t) = \frac{\mathscr{E}^t\{e^{\zeta_t} \mid x_t\}\,p(x, t)}{\mathscr{E}^t\{e^{\zeta_t}\}} \triangleq \frac{Q}{P}\,, \tag{6.81}$$

where Q and P are the numerator and denominator, respectively. We see that Q is a function of ζ_t and, through $p(x, t)$, a function of t. Applying Itô's Lemma 4.2, we get

$$dQ = Q_t\,dt + Q_\zeta\,d\zeta_t + \tfrac{1}{2}h_t^{\mathrm{T}}R^{-1}(t)\,h_t Q_{\zeta\zeta}\,dt.$$

But

$$Q_t = \mathscr{E}^t\{e^{\zeta_t} \mid x_t\}\,\frac{\partial p(x, t)}{\partial t} = \mathscr{L}(Q),$$

and

$$Q_\zeta = Q_{\zeta\zeta} = Q.$$

Therefore,

$$\begin{aligned} dQ &= \mathscr{L}(Q)\,dt + Qh_t^{\mathrm{T}}R^{-1}(t)\,h_t\,dt + Qh_t^{\mathrm{T}}R^{-1}(t)\,d\eta_t \\ &= \mathscr{L}(Q)\,dt + Qh_t^{\mathrm{T}}R^{-1}(t)\,dz_t\,. \end{aligned} \tag{6.82}$$

Now

$$P = \int Q\,dx,$$

so that

$$
\begin{aligned}
dP &= \int (dQ)\, dx = P \int (P^{-1}\, dQ)\, dx \\
&= P\left[\int \mathscr{L}(QP^{-1})\, dt\, dx + \int QP^{-1}h_t^{\mathrm{T}}R^{-1}(t)\, dz_t\, dx\right] \\
&= P\left[\left\{\int \mathscr{L}(QP^{-1})\, dx\right\} dt + \left\{\int QP^{-1}h_t^{\mathrm{T}}R^{-1}(t)\, dx\right\} dz_t\right] \\
&= P\left[\mathscr{L}\left\{\int QP^{-1}\, dx\right\} dt + \left\{\int QP^{-1}h_t^{\mathrm{T}}R^{-1}(t)\, dx\right\} dz_t\right] \\
&= P\mathscr{L}(1)\, dt + P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, dz_t = 0 + P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, dz_t\,,
\end{aligned}
$$

that is,

$$
dP = P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, dz_t = P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, h_t\, dt + P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, d\eta_t\,. \tag{6.83}
$$

Now, in Corollary 3 of Lemma 4.2, let $\varphi = x = Q$ and $\psi = y = P$. Then

$$
\begin{aligned}
d(QP^{-1}) &= P^{-1}\, dQ - QP^{-2}\, dP - P^{-2}Qh_t^{\mathrm{T}}R^{-1}(t)\, R(t)\, R^{-1}(t)\, \hat{h}_t P\, dt \\
&\quad + QP^{-3}P\hat{h}_t^{\mathrm{T}}R^{-1}(t)\, R(t)\, R^{-1}(t)\, \hat{h}_t P\, dt \\
&= \mathscr{L}(QP^{-1})\, dt + (QP^{-1})\, h_t^{\mathrm{T}}R^{-1}(t)\, dz_t - (QP^{-1})\, \hat{h}_t^{\mathrm{T}}R^{-1}(t)\, dz_t \\
&\quad - (QP^{-1})\, h_t^{\mathrm{T}}R^{-1}(t)\, \hat{h}_t\, dt + (QP^{-1})\, \hat{h}_t^{\mathrm{T}}R^{-1}(t)\, \hat{h}_t\, dt \\
&= \mathscr{L}(QP^{-1})\, dt + (QP^{-1})(h_t - \hat{h}_t)^{\mathrm{T}}\, R^{-1}(t)(dz_t - \hat{h}_t\, dt),
\end{aligned}
$$

which is (6.79). ■

Example 6.3. Instead of using the characterization of the conditional density in Lemma 6.2 and Itô's calculus, we compute dp via an expansion procedure similar to that employed in Section 4.2. Terms involving $d\eta_t\, d\eta_t^{\mathrm{T}}$ are replaced by their expectation (see the demonstration of Lemma 4.2).

We wish to compute the change in $p(x, t \mid Y_t)$ due to the dynamics and the differential observation dz_t. Now

$$
\begin{aligned}
\delta p &= p(x, t + \delta t \mid Y_t\,, \delta z_t) - p(x, t \mid Y_t) \\
&= p(x, t + \delta t \mid Y_t\,, \delta z_t) - p(x, t \mid Y_t\,, \delta z_t) + p(x, t \mid Y_t\,, \delta z_t) - p(x, t \mid Y_t) \\
&\triangleq \delta p_d + \delta p_0\,,
\end{aligned} \tag{6.84}
$$

where δp_d is the change due to the dynamics and δp_0 is the change due to the differential observation δz_t. δp_d and δp_0 are independent, since $\{\beta_t\}$ and $\{\eta_t\}$ are assumed independent. Now dp_d has already been computed, and is given by Kolmogorov's forward equation:

$$dp_d = \mathscr{L}(p)\, dt.$$

It remains to compute dp_0.

By Theorem 6.1,

$$p(x, t \mid Y_t, \delta z_t) = \frac{p(\delta z_t \mid x_t)\, p(x, t \mid Y_t)}{\int p(\delta z_t \mid x_t)\, p(x, t \mid Y_t)\, dx}, \tag{6.85}$$

where

$$p(\delta z_t \mid x_t) = C \exp\{(-\tfrac{1}{2}\, \delta t)[\delta z_t - h(x, t)\, \delta t]^{\mathrm{T}} R^{-1}(t)[\delta z_t - h(x, t)\, \delta t]\}. \tag{6.86}$$

Substituting (6.86) into (6.85),

$$p(x, t \mid Y_t, \delta z_t) = \frac{p(x, t \mid Y_t) \exp[\delta z_t{}^{\mathrm{T}} R^{-1}(t)\, h(x, t) - \tfrac{1}{2} h^{\mathrm{T}}(x, t)\, R^{-1}(t)\, h(x, t)\, \delta t]}{\int [\text{numerator}]\, dx}. \tag{6.87}$$

Define

$$E(\delta t, \delta z_t) = \frac{p(x, t \mid Y_t, \delta z_t)}{p(x, t \mid Y_t)}, \tag{6.88}$$

and expand $E(\delta t, \delta z_t)$ in Taylor series about (0, 0), retaining terms of order δt. Since

$$\begin{aligned} \mathscr{E}\{\delta z_t\, \delta z_t{}^{\mathrm{T}}\} &= \mathscr{E}\{h_t h_t{}^{\mathrm{T}}\}\, \delta t^2 + \mathscr{E}\{\delta \eta_t\, \delta \eta_t{}^{\mathrm{T}}\} \\ &= R(t)\, \delta t + o(\delta t), \end{aligned} \tag{6.89}$$

the expansion must be carried out to second degree in δz_t and to first degree in δt. $\delta z_t\, \delta t = o(\delta t)$ in root mean square, so that the mixed terms need not be retained.

It is easy to compute

$$E(0, 0) = 1,$$

$$E_{\delta t}(0, 0) = -\tfrac{1}{2}[h^{\mathrm{T}} R^{-1} h - (h^{\mathrm{T}} R^{-1} h)\hat{}\,], \qquad \text{[see (6.91)]},$$

$$E_{\delta z_t}(0, 0) = R^{-1}(h - \hat{h}),$$

$$E_{\delta z_t, \delta z_t}(0, 0) = R^{-1} h h^{\mathrm{T}} R^{-1} - 2R^{-1} h \hat{h}^{\mathrm{T}} R^{-1} + 2R^{-1} \hat{h} \hat{h}^{\mathrm{T}} R^{-1} - R^{-1} \widehat{h h^{\mathrm{T}}} R^{-1},$$

so that

$$E(\delta t, \delta z_t) = 1 - \tfrac{1}{2}[h^{\mathrm{T}}R^{-1}h - (h^{\mathrm{T}}R^{-1}h)\hat{}\,]\,\delta t + (h - \hat{h})^{\mathrm{T}}\,R^{-1}\,\delta z_t + \tfrac{1}{2}\delta z_t{}^{\mathrm{T}}[R^{-1}hh^{\mathrm{T}}R^{-1} - 2R^{-1}h\hat{h}^{\mathrm{T}}R^{-1} + 2R^{-1}\hat{h}\hat{h}^{\mathrm{T}}R^{-1} - R^{-1}\widehat{hh^{\mathrm{T}}}R^{-1}]\,\delta z_t + r,$$

where

$$\mathscr{E}\{r\} = o(\delta t), \qquad \mathscr{E}\{r^2\} = o(\delta t^2).$$

Replacing terms containing $\delta z_t\,\delta z_t{}^{\mathrm{T}}$ by their expectations, we get

$$E(\delta t, \delta z_t) = 1 + (h - \hat{h})^{\mathrm{T}}\,R^{-1}(\delta z_t - \hat{h}\,\delta t) + r,$$

and, in the mean square sense,

$$dp_0 = (h - \hat{h})^{\mathrm{T}}\,R^{-1}(dz_t - \hat{h}\,dt)\,p. \quad \blacktriangle \tag{6.90}$$

9. EVOLUTION OF MOMENTS (CONTINUOUS)

Let $\varphi(x)$ be a twice continuously differentiable scalar function of the n-vector x. Using Kushner's equation (6.79), we develop a stochastic differential equation for

$$\hat{\varphi}^t(x_t) = [\varphi(x_t)]\hat{}^t = \mathscr{E}^t\{\varphi(x_t)\} = \mathscr{E}\{\varphi(x_t) \mid Y_t\}. \tag{6.91}$$

For convenience of notation, we omit the superscript t and write simply $\hat{\varphi}(x_t)$.

By definition,

$$\hat{\varphi}(x_t) = \int \varphi(x)\,p(x, t \mid Y_t)\,dx,$$

so that

$$\begin{aligned} d\hat{\varphi} &= \int \varphi(x)\,(dp)\,dx \\ &= \left[\int \varphi(x)\,\mathscr{L}(p)\,dx\right] dt + \int \varphi(x)(h_t - \hat{h}_t)^{\mathrm{T}}\,R^{-1}(t)(dz_t - \hat{h}_t\,dt)\,p\,dx \\ &\triangleq I_1 + I_2\,, \end{aligned} \tag{6.92}$$

where we have used (6.79). From Lemma 6.1,

$$I_1 = \widehat{\varphi_x{}^{\mathrm{T}}f}\,dt + \tfrac{1}{2}\,\mathrm{tr}\,(GQG^{\mathrm{T}}\varphi_{xx})\hat{}\,dt, \tag{6.93}$$

and we easily evaluate

$$I_2 = (\widehat{\varphi h} - \hat{\varphi}\hat{h})^{\mathrm{T}} R^{-1}(dz_t - \hat{h}\, dt). \tag{6.94}$$

To summarize, we have

Lemma 6.3. *Hypotheses of Lemma* 6.2. *Let* $\varphi(x)$ *be a twice continuously differentiable scalar function of the n-vector* x. *Then* $\hat{\varphi}(x_t) = \mathscr{E}\{\varphi(x_t) | Y_t\}$ *satisfies*

$$d\hat{\varphi}(x_t) = [\widehat{\varphi_x{}^{\mathrm{T}} f} + \tfrac{1}{2} \operatorname{tr} (GQG^{\mathrm{T}}\varphi_{xx})\hat{}\,]\, dt + (\widehat{\varphi h} - \hat{\varphi}\hat{h})^{\mathrm{T}} R^{-1}(t)(dz_t - \hat{h}\, dt), \qquad t \geqslant t_0 . \tag{6.95}$$

Equation (6.95) has been established only formally, inasmuch as it was implicitly assumed that the conditional density is sufficiently differentiable and well behaved at infinity. Under a multitude of technical conditions, Kushner [10] proves (6.95) rigorously, but we cannot pursue that proof here. Note that, with $R^{-1}(t) \equiv 0$ (valueless or no observations), (6.95) reduces to (6.21) of Lemma 6.1.

Lemma 6.3 can be used to determine (stochastic) differential equations for all the moments of $p(x, t \mid Y_t)$. We now develop the equations for the conditional mean and the conditional covariance matrix. The development is similar to Theorem 6.2, except that here we must use Itô's stochastic calculus.

Setting $\varphi(x) = x_i$, the ith component of the vector x, we immediately get the equations for $\hat{x}_t$:

$$d\hat{x}_i = \hat{f}_i\, dt + (\widehat{x_i h} - \hat{x}_i\hat{h})^{\mathrm{T}} R^{-1}(t)\,(dz_t - \hat{h}\, dt), \qquad i = 1,\ldots, n. \tag{6.96}$$

Now

$$P_t = \mathscr{E}\{(x_t - \hat{x}_t)(x_t - \hat{x}_t)^{\mathrm{T}} \mid Y_t\} = \widehat{x_t x_t{}^{\mathrm{T}}} - \hat{x}_t\hat{x}_t{}^{\mathrm{T}},$$

so that

$$dP_t = d(\widehat{x_t x_t{}^{\mathrm{T}}}) - d(\hat{x}_t\hat{x}_t{}^{\mathrm{T}}). \tag{6.97}$$

Setting $\varphi(x) = x_i x_j$ in (6.95), we have

$$d(\widehat{x_i x_j}) = (\widehat{x_i f_j} + \widehat{f_i x_j} + (GQG^{\mathrm{T}})_{ij}\hat{}\,)\, dt + (\widehat{x_i x_j h} - \widehat{x_i x_j}\hat{h})^{\mathrm{T}} R^{-1}(t)\,(dz_t - \hat{h}\, dt). \tag{6.98}$$

Now $\hat{x}_t$ is the solution of the stochastic differential equation (6.96). Therefore, by Itô's Lemma 4.2,

$$d(\hat{x}_i\hat{x}_j) = \hat{x}_i\, d\hat{x}_j + (d\hat{x}_i)\,\hat{x}_j + (\widehat{x_i h} - \hat{x}_i\hat{h})^{\mathrm{T}} R^{-1}(t)(\widehat{hx_j} - \hat{h}\hat{x}_j)\, dt. \tag{6.99}$$

Substituting from (6.96) for $d\hat{x}_i$, combining (6.99), (6.98), and (6.97), we get the desired equation for dP_t.

Theorem 6.6 (Conditional Mean and Covariance Matrix for Continuous Problems). *Hypotheses of Lemma* 6.2. *Then the conditional mean $\hat{x}_t$ and conditional covariance matrix P_t satisfy the stochastic differential equations*

$$d\hat{x}_t = \hat{f}(x_t, t)\,dt + (\widehat{x_t h^{\mathrm{T}}} - \hat{x}_t \hat{h}^{\mathrm{T}})\,R^{-1}(t)\,(dz_t - \hat{h}\,dt), \tag{6.100}$$

$$\begin{aligned}(dP_t)_{ij} = {} & [(\widehat{x_i f_j} - \hat{x}_i \hat{f}_j) + (\widehat{f_i x_j} - \hat{f}_i \hat{x}_j) + (GQG^{\mathrm{T}})\hat{}_{ij} \\ & - (\widehat{x_i h} - \hat{x}_i \hat{h})^{\mathrm{T}}\,R^{-1}(t)(\widehat{h x_j} - \hat{h}\hat{x}_j)]\,dt \\ & + (\widehat{x_i x_j h} - \widehat{x_i x_j}\hat{h} - \hat{x}_i \widehat{x_j h} - \hat{x}_j \widehat{x_i h} + 2\hat{x}_i \hat{x}_j \hat{h})^{\mathrm{T}}\,R^{-1}(t)\,(dz_t - \hat{h}\,dt).\end{aligned} \tag{6.101}$$

The remarks made in Section 6.4 concerning Theorem 6.2 apply to Eqs. (6.100) and (6.101) as well. Approximations to these equations are considered in Chapter 9. They are specialized to the linear problem in Chapter 7.

10. EVOLUTION OF THE MODE (CONTINUOUS)

We develop here the stochastic differential equation for the mode of $p(x, t \mid Y_t)$ (6.67). The derivation is similar in principle to that of Section 5 except that we use Kushner's equation (6.79) instead of Kolmogorov's forward equation. We *assume* that the mode satisfies a stochastic differential equation and use the formal rules of Itô's stochastic calculus. The derivation is based in part on the work of Kushner [11].

We will need the following formal result from Itô's stochastic calculus.

Lemma 6.4. *Suppose that, for a fixed vector m, x_t is the unique solution of the Itô equation*

$$dx_i = f_i(x, m, t)\,dt + \sum_j G_{ij}(x, m, t)\,d\beta_j\,, \tag{6.102}$$

where $\mathcal{E}\{d\beta_t\,d\beta_t^{\mathrm{T}}\} = Q(t)\,dt$, and where all the functions are assumed to be sufficiently differentiable with respect to all arguments. Now, if m_t is the unique solution of the Itô equation

$$dm_i = \varphi_i(x, m, t)\,dt + \sum_j \Gamma_{ij}(x, m, t)\,d\beta_j\,, \tag{6.103}$$

then dx_t (6.102) *is the partial (time) differential. The total differential of* x_t *is given by*

$$dx_i = \left[f_i + \sum_j (x_i)_{m_j}\varphi_j + \sum_{j,k,l} (G_{ik})_{m_j}\Gamma_{jl}Q_{kl} + \frac{1}{2}\sum_{j,k} (\Gamma Q\Gamma^{\mathrm{T}})_{jk}\,(x_i)_{m_j m_k}\right] dt$$

$$+ \sum_j G_{ij}\, d\beta_j + \sum_{j,k} (x_i)_{m_j}\Gamma_{jk}\, d\beta_k\,. \tag{6.104}$$

In this notation, $(x_i)_{m_j}$ is the partial derivative of x_i with respect to m_j, and so on. $(x_i)_m$ will be the gradient (column) vector of x_i with respect to m, $(x_i)_{mm}$ will be the matrix of second partials, and x_m is the matrix whose rows are the gradients with respect to m. x_i and its partials are evaluated at (m_t, t).

Lemma 6.4 is the stochastic generalization of the analogous deterministic rule that was already used in (6.29) and (6.33) of Section 5. We give now a formal demonstration of the lemma using the expansion procedure used in the demonstration of Lemma 4.2. The expansion must be carried to second order in differentials involving $d\beta_t$, and $d\beta_t\, d\beta_t^{\mathrm{T}}$ is replaced by its expectation $Q(t)\, dt$.

Demonstration. The total differential of x_t is

$$dx_i = x_i(m + dm, t + dt) - x_i(m, t)$$

$$= x_i(m + dm, t + dt) - x_i(m, t + dt) + x_i(m, t + dt) - x_i(m, t).$$

Now

$$x_i(m, t + dt) - x_i(m, t) = dx_i$$

and

$$x\,(m + dm, t + dt) - x_i(m, t + dt)$$

$$= \sum_j (x_i(m, t + dt))_{m_j}\, dm_j + \frac{1}{2}\sum_{j,k} (x_i(m, t + dt))_{m_j m_k}\, dm_j\, dm_k + o(dt).$$

But

$$\sum_j (x_i(m, t + dt))_{m_j}\, dm_j = \sum_j [(x_i(m, t))_{m_j} + d(x_i)_{m_j}]\, dm_j$$

and

$$\frac{1}{2}\sum_{j,k} (x_i(m, t + dt))_{m_j m_k}\, dm_j\, dm_k = \frac{1}{2}\sum_{j,k} (x_i(m, t))_{m_j m_k}\, dm_j\, dm_k + o(dt).$$

Therefore,

$$dx_t = dx_i + \sum_j [(x_t)_{m_j} + d(x_t)_{m_j}]\, dm_j + \frac{1}{2}\sum_{j,k} (x_i)_{m_j m_k}\, dm_j\, dm_k + o(dt). \tag{6.105}$$

From (6.102),

$$d(x_i)_{m_j} = (f_i)_{m_j}\,dt + \sum_k (G_{ik})_{m_j}\,d\beta_k\,. \tag{6.106}$$

Substituting this, (6.102), and (6.103) into (6.105), we get

$$\begin{aligned} \bar{d}x_i = &\left[f_i + \sum_j (x_i)_{m_j}\varphi_j\right] dt + \sum_j G_{ij}\,d\beta_j + \sum_{j,k} (x_i)_{m_j}\Gamma_{jk}\,d\beta_k \\ &+ \sum_{j,k,l} (G_{ik})_{m_j}\Gamma_{jl}\,d\beta_k\,d\beta_l + \frac{1}{2}\sum_{j,k,l,q} (x_i)_{m_j m_k}\Gamma_{jl}\Gamma_{kq}\,d\beta_l\,d\beta_q + o(dt). \end{aligned} \tag{6.107}$$

Replacing $d\beta_i\,d\beta_j$ by its expectation $Q_{ij}\,dt$ produces (6.104).

Now we proceed to the derivation of the equation for the mode. If m_t is the mode of $p(x, t \mid Y_t)$, then, by definition,

$$p_m(m, t \mid Y_t) \equiv 0, \tag{6.108}$$

and consequently

$$\bar{d}p_m(m, t \mid Y_t) = 0. \tag{6.109}$$

Now, from Kushner's equation (6.79),

$$dp_m = (\mathscr{L}(p))_m\,dt + h_m{}^{\mathrm{T}}R^{-1}(dz_t - \hat{h}\,dt)\,p + p_m(h - \hat{h})^{\mathrm{T}}\,R^{-1}\,(dz_t - \hat{h}\,dt), \tag{6.110}$$

or, equivalently,

$$\begin{aligned} dp_m = &[(\mathscr{L}(p))_m + ph_m{}^{\mathrm{T}}R^{-1}(h - \hat{h}) + p_m(h - \hat{h})^{\mathrm{T}}\,R^{-1}(h - \hat{h})]\,dt \\ &+ [ph_m{}^{\mathrm{T}}R^{-1} + p_m(h - \hat{h})^{\mathrm{T}}\,R^{-1}]\,d\eta. \end{aligned} \tag{6.110a}$$

Suppose m_t satisfies (6.103) with β_t replaced by η_t. Then, by Lemma 6.4 (with $x = p_m$),

$$\begin{aligned} 0 = \bar{d}p_{m_i} = &\Big\{[(\mathscr{L}(p))_m + ph_m{}^{\mathrm{T}}R^{-1}(h - \hat{h})]_i + \sum_j p_{m_i m_j}\varphi_j + p\sum_{j,k} (h_k)_{m_i m_j}\Gamma_{jk} \\ &+ \sum_{j,k} p_{m_i m_j}(h - \hat{h})_k\,\Gamma_{jk} + \frac{1}{2}\sum_{j,k} (\Gamma R\Gamma^{\mathrm{T}})_{jk}\,p_{m_i m_j m_k}\Big\}\,dt \\ &+ p\sum_j (h_m{}^{\mathrm{T}}R^{-1})_{ij}\,d\eta_j + \sum_{j,k} p_{m_i m_j}\Gamma_{jk}\,d\eta_k\,, \end{aligned} \tag{6.111}$$

where we have used (6.108).

Now the coefficients of dt and $d\eta$ in (6.111) must be zero. Setting the coefficient of $d\eta$ equal to zero produces

$$\Gamma = -pp_{mm}^{-1}h_m{}^{\mathrm{T}}R^{-1}. \tag{6.112}$$

Putting this into the zero coefficient of dt gives

$$\varphi = -p_{mm}^{-1}(\mathscr{L}(p))_m + p^2 p_{mm}^{-1} u - \tfrac{1}{2} p^2 p_{mm}^{-1} s, \tag{6.113}$$

where u and s are vectors with components

$$u_i = \sum_{j,k} (h_k)_{m_i m_j} (p_{mm}^{-1} h_m{}^{\mathrm{T}} R^{-1})_{jk}, \tag{6.114}$$

$$s_i = \operatorname{tr} p_{mm}^{-1} h_m{}^{\mathrm{T}} R^{-1} h_m p_{mm}^{-1} (p_{m_i})_{mm}. \tag{6.115}$$

Therefore, the mode m_t satisfies (6.103):

$$\begin{aligned} dm_t = {} & [-p_{mm}^{-1}(\mathscr{L}(p))_m + p^2 p_{mm}^{-1} u - \tfrac{1}{2} p^2 p_{mm}^{-1} s]\, dt \\ & - p p_{mm}^{-1} h_m{}^{\mathrm{T}} R^{-1}(t)\, (dz_t - h\, dt), \end{aligned} \tag{6.116}$$

where p and all its partials are evaluated at (m_t, t). We have, of course, assumed that $p_{mm} > 0$. This is the desired stochastic differential equation for the conditional mode.

The right-hand side of (6.116) depends on p, $p_{m_i m_j}$ and $p_{m_i m_j m_k}$. Equations for these derivatives can be derived, and these will depend on still higher-order derivatives. Thus, in order to obtain a practical, realizable maximum likelihood filter, these equations have to be truncated or the partials have to be approximated in some way.

Since in the linear-Gaussian case

$$\Sigma_t = -p p_{mm}^{-1} \tag{6.117}$$

is the conditional covariance matrix, we will derive a stochastic differential equation for it. Because of the tedious algebra, we limit ourselves to the scalar case ($n = m = 1$).

By Lemma 6.4 ($x = p$),

$$\begin{aligned} \bar{d}p &= dp - \tfrac{1}{2}\Sigma_t^2 p_{mm} h_m{}^2 R^{-1}\, dt \\ &= [\mathscr{L}(p) + p(h - \hat{h})^2 R^{-1} - \tfrac{1}{2}\Sigma_t^2 p_{mm} h_m{}^2 R^{-1}]\, dt + p(h - \hat{h}) R^{-1}\, d\eta, \end{aligned} \tag{6.118}$$

where we have used (6.108). From (6.110),

$$dp_{mm} = (\mathscr{L}(p))_{mm}\, dt + (p h_{mm} R^{-1} + p_{mm}(h - \hat{h}) R^{-1} + 2 p_m h_m R^{-1})\, (dz_t - \hat{h}\, dt), \tag{6.119}$$

and, using Lemma 6.4 (with $x = p_{mm}$),

$$\begin{aligned}
\bar{d}p_{mm} &= dp_{mm} + p_{mmm}\,dm_t \\
&\quad + (ph_{mmm} + p_{mmm}(h - \hat{h}) + 3p_{mm}h_m + \tfrac{1}{2}\Sigma_t h_m p_{mmmm})\, h_m R^{-1}\Sigma_t\,dt \\
&= [(\mathscr{L}(p))_{mm} + (ph_{mm} + p_{mm}(h - \hat{h}))\, R^{-1}(h - \hat{h}) \\
&\quad + p_{mmm}(-p_{mm}^{-1}(\mathscr{L}(p))_m + \Sigma_t^2 h_m h_{mm} R^{-1} - \tfrac{1}{2}\Sigma_t^2 R^{-1} h_m^2 p_{mm}^{-1} p_{mmm}) \\
&\quad + (ph_{mmm} + p_{mmm}(h - \hat{h}) + 3p_{mm}h_m + \tfrac{1}{2}\Sigma_t h_m p_{mmmm})\, h_m R^{-1}\Sigma_t]\,dt \\
&\quad + [ph_{mm} + p_{mm}(h - \hat{h}) + \Sigma_t h_m p_{mmm}]\, R^{-1}\,d\eta, \qquad (6.120)
\end{aligned}$$

where again we have used (6.108). Now applying Corollary 3 of Lemma 4.2 with $\varphi = x = p$, $\psi = y = p_{mm}$, we finally get

$$\begin{aligned}
-\bar{d}\Sigma_t &= \bar{d}(pp_{mm}^{-1}) \\
&= p_{mm}^{-1}\,\bar{d}p + \Sigma_t p_{mm}^{-1}\,\bar{d}p_{mm} \\
&\quad + [-\Sigma_t h_{mm}^2 + h_{mm}(h - \hat{h}) + 2\Sigma_t p_{mm}^{-1} h_m h_{mm} p_{mmm} \\
&\quad - p_{mm}^{-1}(h - \hat{h})\, h_m p_{mmm} - \Sigma_t p_{mm}^{-2} h_m^2 p_{mmm}^2]\, \Sigma_t^2 R^{-1}\,dt. \qquad (6.121)
\end{aligned}$$

Substituting (6.118) and (6.120) into (6.121) yields the desired result.

We summarize these results in the following theorem.

Theorem 6.7 (Conditional Mode for Continuous Problems). *Hypotheses of Lemma* 6.2. *Let* $p(x, t \mid Y_t)$ *be sufficiently differentiable and* $p_{mm}(m_t, t \mid Y_t) > 0$. *Suppose the conditional mode* m_t *satisfies an (Itô) stochastic differential equation. Then the conditional mode satisfies*

$$dm_t = [-p_{mm}^{-1}(\mathscr{L}(p))_m - p\Sigma_t(u - \tfrac{1}{2}s)]\,dt + \Sigma_t h_m{}^{\mathrm{T}} R^{-1}(t)\,(dz_t - h\,dt), \qquad (6.122)$$

where the n-vectors u and s are defined in (6.114) *and* (6.115), *and where p, its partials, h, and all other functions are evaluated at* (m_t, t). *In the scalar case* $(n = m = 1)$, $\Sigma_t\,(= -pp_{mm}^{-1})$ *satisfies*

$$\begin{aligned}
\bar{d}\Sigma_t &= -p_{mm}^{-1}[\mathscr{L}(p) + \Sigma_t(\mathscr{L}(p))_{mm} - p_{mmm}\Sigma_t p_{mm}^{-1}(\mathscr{L}(p))_m]\,dt \\
&\quad + \Sigma_t^2(h_{mm} - p_{mm}^{-1}h_m p_{mmm})\, R^{-1}(t)\,(dz_t - \hat{h}\,dt) \\
&\quad - \Sigma_t p_{mm}^{-1}[3p_{mmm}\Sigma_t^2 h_m(h_{mm} - \tfrac{1}{2}h_m p_{mm}^{-1}p_{mmm})\, R^{-1}(t)\,dt \\
&\quad + (ph_{mmm} + \tfrac{5}{2}p_{mm}h_m + \tfrac{1}{2}\Sigma_t h_m p_{mmmm})\, h_m R^{-1}(t)\,\Sigma_t\,dt] \\
&\quad + \Sigma_t^3 h_{mm}^2 R^{-1}(t)\,dt + \Sigma_t^2(h - \hat{h})(p_{mm}^{-1}h_m p_{mmm} - h_{mm})\, R^{-1}(t)\,dt. \qquad (6.123)
\end{aligned}$$

Note that (6.122) and (6.123) reduce to (6.40) and (6.41), respectively, of Theorem 6.3 in case $R^{-1}(t) \equiv 0$. Comparing the equation for the conditional mode with that for the conditional mean (6.100), we see an important qualitative difference. The equation for the mean depends on average properties of the functions f, h, and so on, whereas the equation for the mode depends on local properties of these functions. Only indirectly through Σ_t does the mode depend on average properties. See also Example 6.1.

Example 6.4. In the scalar case ($n = m = 1$),

$$\begin{aligned}
\mathscr{L}(p)(m, t) &= -pf_m + \tfrac{1}{2}(p_{mm}\sigma^2 + p\sigma^2_{mm}),\\
(\mathscr{L}(p))_m(m, t) &= -p_{mm}f - pf_{mm} + \tfrac{1}{2}(p_{mmm}\sigma^2 + 3p_{mm}\sigma_m{}^2 + p\sigma^2_{mmm}),\\
(\mathscr{L}(p))_{mm}(m, t) &= -p_{mmm}f - 3p_{mm}f_m - pf_{mmm}\\
&\quad + \tfrac{1}{2}(p_{mmmm}\sigma^2 + 4p_{mmm}\sigma_m{}^2 + 6p_{mm}\sigma^2_{mm} + p\sigma^2_{mmmm}),
\end{aligned} \tag{6.124}$$

where $\sigma^2 = GQG^{\mathrm{T}}$ and all the functions are evaluated at (m, t).

We specialize (6.122) and (6.123) to the linear-Gaussian, scalar case ($f = F(t)\,x$, $h = M(t)\,x$, $\sigma_x{}^2 \equiv 0$, $p_{mmm} = 0$, $p_{mmmm}/p_{mm} = -3\Sigma_t^{-1}$):

$$\begin{aligned}
dm_t &= F(t)\,m_t\,dt + \Sigma_t M(t)\,R^{-1}(t)\,(dz_t - M(t)\,m_t\,dt),\\
d\Sigma_t/dt &= 2\Sigma_t F(t) + \sigma^2 - \Sigma_t M(t)\,R^{-1}(t)\,M(t)\,\Sigma_t\,.
\end{aligned} \tag{6.125}$$

Equations (6.125) constitute the maximum likelihood (Bayesian) filter in the continuous, linear-Gaussian, scalar case. This is clearly also the minimum variance filter; the mode m_t is the conditional mean, and Σ_t is the conditional variance. Compare this with Examples 6.1 and 6.2. Equations (6.125) will be discussed in some detail in the next chapter. ▲

11. LIMITED MEMORY FILTER

The filters (predictors) developed in the preceding sections utilize all the available data, including all observations and prior data. It is clearly optimal to base an estimate on all the available information. This is predicated, however, on the knowledge of the dynamics, measurement function, and the statistical parameters associated with the dynamical system. If, for example, the dynamics are imprecisely known, then the filter might learn the state "too well." Because of imprecise dynamics used in the filter, the "good estimate" may be wrong.

This problem is treated in detail in Chapter 8. It is referred to as

the problem of filter *divergence*. A reasonable approach to this problem is to limit the filter memory so that the estimate does not become "too good." By limiting the filter memory, we mean computing an estimate based (conditioned) on data from only the recent past. Alternatively, this means discarding the conditioning (of the estimate) on the distant past. The memory length of the filter should coincide with the (time) length over which the approximate dynamical system represents a satisfactory approximation to reality. We develop the theory of the limited memory filter in the present section. The development is based on the work of Jazwinski [6].

Let the dynamical system be discrete (6.58) or continuous (6.64), but without the random forcing term. That is, the state evolves either according to

$$x_{k+1} = \varphi(x_k, t_{k+1}, t_k), \qquad k = 0, 1,..., \tag{6.126D}$$

or according to

$$dx_t/dt = f(x_t, t), \qquad t \geqslant t_0, \tag{6.126C}$$

where x_k is x_{t_k}. Discrete observations (6.59) are taken as

$$y_k = h(x_k, t_k) + v_k, \qquad k = 1, 2,..., \tag{6.127}$$

where $\{v_k\}$ is white with the usual statistics.

Consider the sequence of data

$$y_1,..., y_m, y_{m+1},..., y_n,...; \qquad n - m = N > 0, \tag{6.128}$$

and define

$$\begin{aligned} Y_m &= \{y_1,..., y_m\}, \\ Y_n &= \{y_1,..., y_n\}, \\ Y_N &= \{y_{m+1},..., y_n\}. \end{aligned} \tag{6.129}$$

Lemma 6.5. *Assume dynamical system* (6.126) *with observations* (6.127). *Suppose that the required density functions exist. Then*

$$p(Y_N \mid x_n) = c\,\frac{p(x_n \mid Y_n)}{p(x_n \mid Y_m)}, \tag{6.130}$$

where

$$c = \frac{p(Y_n)}{p(Y_m)}$$

is a constant, independent of x_n. *Also,*

$$p(x_n \mid Y_N) = c_1\,\frac{p(x_n \mid Y_n)}{p(x_n \mid Y_m)}, \tag{6.131}$$

where

$$c_1 = \frac{p(Y_n)}{p(Y_m)\,p(Y_N)}\,p(x_n).$$

PROOF: By Bayes' rule,

$$p(x_n \mid Y_n) = \frac{p(Y_n \mid x_n)\,p(x_n)}{p(Y_n)} = \frac{p(Y_m\,,\,Y_N \mid x_n)\,p(x_n)}{p(Y_n)}.$$

But

$$p(Y_m\,,\,Y_N \mid x_n) = p(Y_m \mid Y_N\,,\,x_n)\,p(Y_N \mid x_n),$$

and, since the dynamical system (6.126) is noise-free,

$$p(Y_m \mid Y_N\,,\,x_n) = p(Y_m \mid x_n).$$

Therefore,

$$p(x_n \mid Y_n) = \frac{p(Y_m \mid x_n)\,p(Y_N \mid x_n)\,p(x_n)}{p(Y_n)}.$$

Now, applying Bayes' rule to $p(Y_m \mid x_n)$,

$$p(Y_m \mid x_n) = \frac{p(x_n \mid Y_m)\,p(Y_m)}{p(x_n)},$$

and rearranging terms,

$$p(x_n \mid Y_n) = \frac{p(Y_m)}{p(Y_n)}\,p(x_n \mid Y_m)\,p(Y_N \mid x_n),$$

which is (6.130). Equation (6.131) follows after application of Bayes' rule to $p(Y_N \mid x_n)$:

$$p(Y_N \mid x_n) = \frac{p(x_n \mid Y_N)\,p(Y_N)}{p(x_n)}. \quad \blacksquare \tag{6.132}$$

Consider the case in which the dynamical system is continuous (6.126C). $p(x_n \mid Y_n)$ is the output of the continuous-discrete filter, and $p(x_n \mid Y_m)$ is the output of the predictor (Theorem 6.1). Then Eq. (6.130) gives the appropriate density for computing the maximum likelihood estimate of x_n based on Y_N.

The conditional density $p(x_n \mid Y_N)$ in (6.131) is the density of x_n given Y_N *and* the density of x_0. It is defined in terms of the joint density $p(x_n\,,\,Y_N)$ and the marginal density $p(Y_N)$ (see Section 2.5). What we wish to do is to define the conditional density of x_n given *only* Y_N, excluding any prior information about x_0. To do this, we suppose

formally that, for each n, $p(x_n)$ is Gaussian with zero-mean and covariance matrix

$$(1/\epsilon)\, I,$$

and *define* $p(x_n \mid Y_N \mid)$, the density of x_n given only Y_N, by

$$p(x_n \mid Y_N \mid) = \lim_{\epsilon \to 0} p(x_n \mid Y_N), \tag{6.133}$$

assuming the limit exists.

Theorem 6.8. *Hypotheses of Lemma* 6.5. *Then*

$$p(x_n \mid Y_N \mid) = c_2 \frac{p(x_n \mid Y_n)}{p(x_n \mid Y_m)}, \tag{6.134}$$

where c_2 is a constant, independent of x_n. [The normalizing constant can obviously be determined by

$$\frac{1}{c_2} = \int \frac{p(x_n \mid Y_n)}{p(x_n \mid Y_m)} dx_n \,.]$$

PROOF: From (6.132),

$$\frac{p(Y_N \mid x_n)}{p(x_n \mid Y_N)} = \frac{p(Y_N)}{p(x_n)}. \tag{6.135}$$

Now $p(Y_N \mid x_n)$ does not depend on the statistics of x_n. Assuming the limit in (6.133) exists,[6] and taking the limit in (6.135), we get

$$\frac{p(Y_N \mid x_n)}{p(x_n \mid Y_N \mid)} = \lim_{\epsilon \to 0} \frac{p(Y_N)}{p(x_n)} = c_3 \,, \tag{6.136}$$

which proves that c_3 exists. c_3 is clearly independent of x_n. (6.136), together with (6.130) of Lemma 6.5, proves the theorem. ■

As was already noted, $p(x_n \mid Y_n)$ and $p(x_n \mid Y_m)$ may be computed via Theorem 6.1. Equation (6.134) then produces the density of x_n given only Y_N. From it, we can compute the moments. In particular, the mean of

$$p(x_n \mid Y_N \mid)$$

is the minimum variance estimate of x_n based on the most recent N observations. In view of (6.136), $p(Y_N \mid x_n)$ and $p(x_n \mid Y_N \mid)$ are equal up to a multiplicative constant. Thus, any estimate of x_n obtained from

[6] This is proved for linear-Gaussian problems in Chapter 7.

$p(Y_N \mid x_n)$ can also be obtained from $p(x_n \mid Y_N \mid)$ by the same operation, and vice versa. We shall return to the limited memory filter in the next chapter, where we will consider its linear version and its mechanization.

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